

Phenix Data Analysis

7 documentation pages

Generated from local Phenix documentation

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Assessing Data Quality

xray-data-quality.html

Why

The first step after data processing is analysis of the diffraction data to assess quality and detect any pathologies that might make structure solution difficult. Data quality can be quantified in terms of resolution of diffraction, anomalous signal-to-noise, and consistency with prior knowledge about diffraction from crystals.

Possible pathologies include twinning, translational non-crystallographic symmetry, anisotropy, and missed symmetry elements. For example, crystals are considered 'twinned' if two or more separate crystals (domains) are intergrown in such a way that they share some crystal lattice points in a symmetrical manner. And translational noncrystallographic symmetry (tNCS) occurs when two or more independent copies of a molecule or assembly have a similar orientation in the asymmetric unit of the crystal.

How

The phenix.xtriage program analyzes data to assess quality and detect possible problems. For most uses, the program only requires a reflection file containing the data set you wish to analyze. Running phenix.xtriage generates information about the data set, which is best viewed in the GUI as graphs and tables. The first section of the output provides an overall assessment of the data.

How to use the phenix.xtriage GUI: [Click here](#)

Phenix reference manual for phenix.xtriage

Common issues

Interpretation of twinning results: There are two parts to testing for twinning — determining whether the overall intensity statistics look normal, and testing for specific twinning operations. Both kinds of tests need to indicate twinning for the data set to be considered twinned. In some cases, the tests of specific twinning operations return a high twin fraction but the overall intensity statistics look normal; this should not be interpreted as twinning. More likely, a symmetry element is missing, and phenix.xtriage might suggest a higher symmetry.

Related programs and Documentation

- **phenix.plan_sad_experiment**: This tool allows you to plan your SAD data collection strategy by estimating the anomalous signal and the probability of solving your structure with SAD based on the anomalously-scattering atoms, wavelength, size of your molecule, and overall $I/\sigma(I)$ for your dataset.
- **phenix.anomalous_signal**: This tool is like phenix.plan_sad_experiment except that it uses your measured data to give much more accurate estimates. Specifically, it uses your scaled data, two half-dataset files of your scaled data, the name of the anomalously-scattering atom, and the number of sites in the substructure in order to estimate the anomalous signal in your data and the probability of solving your structure by SAD. You can create the necessary scaled half-datasets using phenix.scale_and_merge.
- **HKLviewer**: This program visualizes reciprocal space data. This can be very useful for detecting systematic problems, such as incompleteness (e.g., missing cones or wedges), tNCS or twinning.
- **phenix.explore_metric_symmetry**: This program tests whether your crystal lattice is consistent with other lattices/symmetries. It also can test whether two lattices are related. (Command line only).
- **Using unmerged data in Phenix**: You can analyze unmerged data in Phenix to provide useful information, including the new CC1/2 and CC* statistics. The program phenix.merging_statistics

calculates these and a number of other statistics. The program `phenix.scale_and_merge` scales unmerged data for you.

Editing and combining reflection files with the Phenix GUI

reflection_file_editor.html

Contents

- Overview
- Dealing with R-free flags
- Limitations
- Notes on usage

Overview

Phenix includes a simple program for combining reflection data from different files and manipulating R-free flags. Any combination of data in the form of Miller arrays (any set of values indexed by h,k,l) may be merged into a single file, with the limitation that the crystal lattices specified in each file must be compatible. All input formats readable by Phenix are supported, but only MTZ files are written. Output is limited to 25 different columns, which usually corresponds to significantly fewer Miller arrays (for example, experimental phases or Hendrickson-Lattman coefficients usually consist of four columns).

To add an input file, click the "+" button below the top list or simply drag files from the desktop into the list. Available Miller arrays will automatically be loaded into the input list, and the crystal symmetry (if present in the file(s)) will be updated. Select arrays by drag-and-drop into the output list or using the buttons below. If different arrays have the same column label this will be noted in the window. You can edit the output labels and resolution limits for individual arrays by clicking the "Edit arrays. . ." button. You can also manipulate the output data by merging anomalous data or converting to and from intensities and amplitudes.

Dealing with R-free flags

If an R-free set is not present in the output file you may create one by clicking the "R-free generation" button. The default behavior is to flag 5% of reflections up to a maximum of 2000. Currently the assignment of flags is completely random across the entire resolution range, but will default to using the highest possible lattice symmetry initially and expanding to the actual symmetry. (In other words, if the space group is P4, the R-free set will be generated in P422 and expanded to P4.) Alternately, you may pick the test set in thin resolution shells, which helps avoid bias due to non-crystallographic symmetry. (Since the test set will not be evenly distributed across the entire resolution range, don't use this option if NCS is not present.)

An existing R-free set may also be extended to the entire resolution range of the output file, e.g. when switching to a higher-resolution dataset for a partially refined structure. This will also use the highest possible lattice symmetry, and will automatically determine the test set size. If you previously assigned the test set in thin shells using DATAMAN or SFTOOLS, the program will attempt to detect this and issue a warning.

Phenix uses the CNS/XPLOR convention for R-free flags, where 1 marks the test set and 0 the "working" set. By default, all R-free arrays will be written in this format as well, regardless of source; if you generated R-free flags in CCP4, which assigns random numbers (usually 0 through 19) and uses 0 to mark the test set, they will be converted first. If you prefer to preserve the original integer values, the R-free flags dialog includes an option to do this; however, this is incompatible with extending an existing set. Alternately, you may choose to output all R-free flags in the CCP4 format even if extending existing flags, but this will renumber all flags not in the test set. (It also assumes that the test set size is approximately 5% of the total number of reflections.)

Limitations

The editor is primarily designed around common functions operating on a single experimental dataset - in particular the manipulation of R-free flags. It is not suitable for some more complex (but less frequently used) scenarios. A few specific limitations:

Unmerged data is allowed as input, but the output will always be merged (with anomalous pairs preserved if present). Some features of the MTZ format are not supported. In particular, all arrays will be treated as part of the same "crystal" and "dataset" objects in the hierarchy of the MTZ file. This restricts all output arrays to using the same unit cell dimensions. For this reason, the editor may be unsuitable for multi-crystal experiments such as MIR. The wavelength (if defined) will also be the same for all arrays. If not defined it will default to zero.

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- The wavelength (if defined) will also be the same for all arrays. If not defined it will default to zero.

Notes on usage

The editor works best when all input files are in MTZ format; other formats should work but have not been tested as thoroughly. Output labels will be guessed automatically, but this breaks down when merging anomalous data, which halves the number of output arrays (and makes the trailing (+) and (-) unnecessary). This will be checked at runtime. Some data processing programs output the columns F SIGF DANO SIGDANO, which if they occur sequentially in an MTZ file will be treated as a single data array. Internally, these are converted to anomalous data (i.e. F(+) SIGF(+) F(-) SIGF(-)), instead of keeping the original non-anomalous amplitudes and separate anomalous differences. We recommend against using these data in PHENIX, as they are less reliable than the original anomalous amplitudes or intensities (and are not otherwise used here). The editor will not otherwise attempt to impose sensible label names, except when converting between intensities and amplitudes. All input files must either have the same symmetry (allowing for slight differences in unit cell dimensions and screw axes) or leave it undefined (e.g. CNS format). To mix arrays from files with different symmetry, first convert each to the desired final symmetry, then combine them. If you want to convert the data into a higher-symmetry space group based on suggestions from Xtriage, the Xtriage GUI now has an option to do this directly instead of running the editor separately.

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reflection file tools

reflection_file_tools.html

phenix.reflection_file_converter

Purpose

phenix.reflection_file_converter is a simple utility program that allows a straightforward conversion of many reflection file formats to mtz, cns, scalepack or shelx format. Currently, combining several dataset into a single output file is not supported.

Keywords

Typing:

```
phenix.reflection_file_converter --help
```

results in:

```
Usage: phenix.reflection_file_converter [options] reflection_file ...
```

Options:

```
-h, --help show this help message and exit
--unit_cell=10,10,20,90,90,120|FILENAME
External unit cell parameters
--space_group=P212121|FILENAME
External space group symbol
--symmetry=FILENAME External file with symmetry information
--weak_symmetry symmetry on command line is weaker than symmetry found
in files
--resolution=FLOAT High resolution limit (minimum d-spacing, d_min)
--low_resolution=FLOAT
Low resolution limit (maximum d-spacing, d_max)
--label=STRING Substring of reflection data label or number
--non_anomalous Averages Bijvoet mates to obtain a non-anomalous array
--r_free_label=STRING
Substring of reflection data label or number
--r_free_test_flag_value=FLOAT
Value in R-free array indicating assignment to free
set.
--generate_r_free_flags
Generates a new array of random R-free flags (MTZ and
CNS output only).
--use_lattice_symmetry_in_r_free_flag_generation
group twin/pseudo symmetry related reflections
together in r-free set (this is the default).
--no_lattice_symmetry_in_r_free_flag_generation
opposite of --use_lattice_symmetry_in_r_free_flag_
generation
--r_free_flags_fraction=FLOAT
Target fraction free/work reflections (default: 0.10).
--r_free_flags_max_free=INT
Maximum number of free reflections (default: 2000).
--r_free_flags_format=cns|ccp4
Convention for generating R-free flags
--random_seed=INT Seed for random number generator (affects generation
of R-free flags).
--change_of_basis=STRING
Change-of-basis operator: h,k,l or x,y,z or
to_reference_setting, to_primitive_setting,
to_niggli_cell, to_inverse_hand
--eliminate_invalid_indices
Remove indices which are invalid given the change of
```

basis desired

--expand_to_p1 Generates all symmetrically equivalent reflections. The space group symmetry is reset to P1. May be used in combination with --change_to_space_group to lower the symmetry.

--change_to_space_group=SYMBOL|NUMBER
Changes the space group and merges equivalent reflections if necessary

--write_mtz_amplitudes
Converts intensities to amplitudes before writing MTZ format; requires --mtz_root_label

--write_mtz_intensities
Converts amplitudes to intensities before writing MTZ format; requires --mtz_root_label

--remove_negatives Remove negative intensities or amplitudes from the data set

--massage_intensities
'Treat' negative intensities to get a positive amplitude. $|F_{new}| = \sqrt{(I_o + \sqrt{I_o^2 + 2\sigma^2})/2.0}$. Requires intensities as input and the flags --mtz, --write_mtz_amplitudes and --mtz_root_label.

--scale_max=FLOAT Scales data such that the maximum is equal to the given value (ignored for --shelx)

--scale_factor=FLOAT Multiplies data with the given factor (ignored for --shelx)

--sca=FILE write data to Scalepack FILE ('--sca .' copies name of input file)

--mtz=FILE write data to MTZ FILE ('--mtz .' copies name of input file)

--mtz_root_label=STRING
Root label for MTZ file (e.g. Fobs)

--cns=FILE write data to CNS FILE ('--cns .' copies name of input file)

--shelx=FILE write data to SHELX FILE ('--shelx .' copies name of input file). Data are automatically scaled if necessary to fit into 8 column format.

Example: phenix.reflection_file_converter w1.sca --mtz .

Examples

- Convert scalepack into an mtz format. Specify output filename (w1.mtz) and label for intensities (IP -> IP,SIGIP): phenix.reflection_file_converter w1.sca --mtz_root_label=IP --mtz=w1.mtz

Convert scalepack into an mtz format. Specify output filename (w1.mtz) and label for intensities (IP -> IP,SIGIP):

```
phenix.reflection_file_converter w1.sca --mtz_root_label=IP --mtz=w1.mtz
```

- Convert mtz intensities into SHELX .hkl format. Specify output filename (data.hkl):
phenix.reflection_file_converter data.mtz --label=IOBS,SIGIOBS --shelx=data.hkl

Convert mtz intensities into SHELX .hkl format. Specify output filename (data.hkl):

```
phenix.reflection_file_converter data.mtz --label=IOBS,SIGIOBS --shelx=data.hkl
```

- Change basis to get data in primitive setting, merge to higher symmetry and bring to reference setting (three steps):
phenix.reflection_file_converter c2.sca --change_of_basis=to_niggli_cell --sca=niggli.sca
phenix.reflection_file_converter niggli.sca --change_to_space_group=R32:R --sca=r32r.sca
phenix.reflection_file_converter r32r.sca --change_of_basis=to_reference_setting
--sca=r32_hexagonal_setting.sca

Change basis to get data in primitive setting, merge to higher symmetry and bring to reference setting (three steps):


```
phenix.reflection_file_converter c2.sca --change_of_basis=to_niggli_cell --sca=niggli.sca
phenix.reflection_file_converter niggli.sca --change_to_space_group=R32:R --sca=r32r.sca
phenix.reflection_file_converter r32r.sca --change_of_basis=to_reference_setting --sca=r32_hexagonal_setting.sc
```

phenix.cns_as_mtz

Purpose

Converts all data in a CNS reflection file to MTZ format.

Keywords

Typing:

```
phenix.cns_as_mtz --help
```

results in:

```
Usage: phenix.cns_as_mtz [options] cns_file
```

Options:

```
-h, --help show this help message and exit
--unit_cell=10,10,20,90,90,120|FILENAME
External unit cell parameters
--space_group=P212121|FILENAME
External space group symbol
--symmetry=FILENAME External file with symmetry information
-q, --quiet suppress output
```

```
Example: phenix.cns_as_mtz scale.hkl
```

Example

Extract unit cell parameters and space group symbol from a PDB coordinate file and reflection data from a CNS reflection file. Write MTZ file:

```
phenix.cns_as_mtz mad_scale.hkl --symmetry minimize.pdb
```

phenix.mtz.dump

Purpose

Inspects an MTZ file. Optionally writes data in text format (human readable, machine readable, or spreadsheet).

Keywords

Typing:

```
phenix.mtz.dump --help
```

results in:

```
Usage: phenix.mtz.dump [options] file_name [...]
```

Options:

```
-h, --help show this help message and exit
-v, --verbose Enable CMTZ library messages.
-c, --show_column_data
-f KEYWORD, --column_data_format=KEYWORD
Valid keywords are: human_readable, machine_readable,
spreadsheet. Human readable is the default. The format
keywords can be abbreviated (e.g. -f s).
-b, --show_batches
--walk=ROOT_DIR Find and process all MTZ files under ROOT_DIR
```

Reflection Statistics

reflection_statistics.html

phenix.reflection_statistics

Comparisons between multiple datasets are available via the phenix.reflection_statistics command:

```
Usage: phenix.reflection_statistics [options] reflection_file [...]
```

Options:

```
-h, --help show this help message and exit
--unit-cell=10,10,20,90,90,120|FILENAME
External unit cell parameters
--space-group=P212121|FILENAME
External space group symbol
--symmetry=FILENAME External file with symmetry information
--weak-symmetry symmetry on command line is weaker than symmetry found
in files
--quick Do not compute statistics between pairs of data arrays
--resolution=FLOAT High resolution limit (minimum d-spacing, d_min)
--low-resolution=FLOAT
Low resolution limit (maximum d-spacing, d_max)
--bins=INT Number of bins
--bins-twinning-test=INT
Number of bins for twinning test
--bins-second-moments=INT
Number of bins for second moments of intensities
--lattice-symmetry-max-delta=LATTICE_SYMMETRY_MAX_DELTA
angular tolerance in degrees used in the determination
of the lattice symmetry
```

```
Example: phenix.reflection_statistics data1.mtz data2.sca
```

This utility reads one or more reflection files (many common formats incl. MTZ, Scalepack, CNS, SHELX). For each of the datasets found in the reflection files the output shows a block like the following:

```
Miller array info: gere_MAD.mtz:FSEinfl,SIGFSEinfl,DSEinfl,SIGDSEinfl
Observation type: xray.reconstructed_amplitude
Type of data: double, size=20994
Type of sigmas: double, size=20994
Number of Miller indices: 20994
Anomalous flag: 1
Unit cell: (108.742, 61.679, 71.652, 90, 97.151, 90)
Space group: C 1 2 1 (No. 5)
Systematic absences: 0
Centric reflections: 0
Resolution range: 24.7492 2.74876
Completeness in resolution range: 0.873513
Completeness with d_max=infinity: 0.872315
Bijvoet pairs: 10497
Lone Bijvoet mates: 0
Anomalous signal: 0.1065
```

This is followed by a listing of the completeness and the anomalous signal in resolution bins. The number of bins and the resolution range may be adjusted with the options shown above.

Unless the --quick option is specified the output will also show the correlations between the datasets and, if applicable, between the anomalous differences, both as overall values and in bins. The correlation between anomalous differences is often a very powerful indicator for the resolution up to which the anomalous signal is useful for substructure determination.

See also: phenix.xtriage

Generating publication content with phenix.table_one

table_one.html

This program is experimental and still in development! Please report any problems to help@phenix-online.org.

Overview

The utility phenix.table_one is a tool for generating the standard table of crystallographic statistics required by most scientific journals. It combines the validation features of phenix.model_vs_data with the ability to calculate merging statistics normally only found in data processing log files. Multiple structures may be input, and a separate column will be generated for each one.

The program is available both in the GUI and the command line; however, because the multiple-structure input makes the command line version more cumbersome to run, this document will generally reference the GUI version. A full list of parameters (with brief explanations) can be found below.

Configuration

Each structure has a separate tab for data input. The final model and data file including R-free flags are required; these should usually be the same files you will submit to the PDB. It is recommended that you provide any CIF (restraints) files for non-standard ligands present in the model; otherwise, eLBOW will be run to generate restraints for any unrecognized residues, but this is frequently less accurate when only the model is available. If you have multiple CIF files in a single directory you may specify the directory path instead of the individual files.

If you have the scaled-but-unmerged intensities (for instance, processed using the "no merge original index" macro in HKL2000/Scalepack), you can also enter these at the bottom of the panel. This will allow the program to re-calculate merging statistics such as the multiplicity/redundancy, R-sym, and R-meas. (Previous versions also attempted to parse the logfiles from data processing software, but this feature has been removed due to unreliability and difficulty of maintenance.) If you do not supply unmerged data, the fields relevant to this step will be left blank in the final document.

The program has relatively few other options beyond the structure inputs. Multiple processors may be utilized if you have input more than one dataset. The most important option is the choice of whether to use phenix.model_vs_data to calculate the final R-factors, or take the values reported in the PDB header. If the model was refined in phenix.refine and the full resolution range present in the data file was used, these should be nearly identical. For other programs such as REFMAC, there is typically a small margin of error (usually around one percent or less) due to implementation details. In addition, for structures with small twin fractions which did not use a twin target in refinement, phenix.model_vs_data may recalculate R-factors that are several percent lower. In these cases it is better to use the values reported in the header, to avoid discrepancy between the article and the PDB deposition.

Output files

By default the program outputs text and RTF formats in a new directory. The RTF file may be edited with Microsoft Word or most other word processing software, and/or copied into another document. Final appearance of a typical result is shown below; usually some editing is required to ensure proper formatting. For data-related statistics (including the R-factors), values for the highest-resolution 10% of reflections will be displayed in parentheses next to the overall value.

List of all available keywords

{{phil:phenix.table_one}}

Data quality assessment with phenix.xtriage

xtriage.html

Contents

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- Literature
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- List of all available keywords

Author(s)

- xtriage: Peter Zwart
- Phil command interpreter: Ralf W. Grosse-Kunstleve

Purpose

The xtriage method is a tool for analyzing structure factor data to identify outliers, presence of twinning and other conditions that the user should be aware of.

Usage

How xtriage works

Basic sanity checks performed by xtriage are

- Wilson plot sanity
- Probabilistic Matthews analysis
- Data strength analysis
- Ice ring analysis
- Twinning analysis

- Reference analysis (determines possible re-indexing. optional)
- Detwinning and data massaging (optional)

See also: ``phenix.reflection_statistics <reflection_statistics.html>`__` (comparison of multiple data sets)

Output files from xtriage

- (1) A log file that contains all the screen output plus some ccp4 style graphs
- optional: an mtz file with massaged data
- optional: an mtz file with massaged data

Key items to look for in Xtriage output

- You want to know if your data has any serious problems such as a large number of implausible values, a large amount of missing data or high values in regions corresponding to ice rings.
- You need to know if your data indicate translational non-crystallographic symmetry.
- You need to know if your data indicate twinning.
- You want to know the most likely space group of the data.

Key actions based on Xtriage output

- If your data has serious problems, you may need to reprocess it or recollect it.
- If there is translational non-crystallographic symmetry, you will want to check to make sure that is included in molecular replacement and refinement.
- If twinning is indicated, you will want to examine your space group, non-crystallographic symmetry, and twinning very carefully, as these characteristics have overlapping effects on the data. Note that you cannot use R values from refinement to determine whether a twin law is correct because refinement with a twin law can lower R values even if there is no twinning. Normally you want to use the statistics of the data to identify the space group, non-crystallographic symmetry, and twinning.
- If the space group is uncertain, work with the most likely one, but keep all the possibilities in mind as you go ahead in case you need to backtrack and try another one. Note that pairs of space groups that are enantiomeric (mirror images, such as P61 and P65) cannot be distinguished from the data alone, so you have to try both when determining your structure. Sets of space groups in the same point group (same rotations, different translations) can be distinguished by analyzing systematic absences, however if the reflections that might show systematic absences are not measured, these space groups might not be distinguishable based on the data alone and you would have to try more than one.

Xtriage keywords in detail

Scope: `parameters.asu_contents`

```
keys: * n_residues :: Number of residues per monomer/unit
      * n_bases :: Number of nucleotides per monomer/unit
      * n_copies_per_asu :: Number of copies in the ASU.
```

These keywords control the determination of the absolute scale. If the number of residues/bases is not specified, a solvent content of 50% is assumed.

Scope: `parameters.misc_twin_parameters.missing_symmetry`

```
keys: * tanh_location :: tanh decision rule parameter
```

```
* tanh_slope :: tanh decision rule parameter
```

The tanh_location and tanh_slope parameter control what R-value is considered to be low enough to be considered a 'proper' symmetry operator. the tanh_location parameter corresponds to the inflection point of the approximate step function. Increasing tanh_location will result in large R-value thresholds. tanh_slope is set to 50 and should be okay.

Scope: parameters.misc_twin_parameters.twinning_with_ncs

```
keys: * perform_test :: can be set to True or False
* n_bins :: Number of bins in determination of D_ncs
```

The perform_test is by default set to False. Setting it to True triggers the determination of the twin fraction while taking into account NCS parallel to the twin axis.

Scope: parameters.misc_twin_parameters.twin_test_cuts

```
keys: * high_resolution : high resolution for twin tests
* low_resolution: low resolution for twin tests
* isigi_cut: I/sig(I) threshold in automatic determination
of high resolution limit
* completeness_cut: completeness threshold in automatic
determination of high resolution limit
```

The automatic determination of the resolution limit for the twinning test is determined on the basis of the completeness after removing intensities for which $I/\sigma(I) < \text{isigi_cut}$. The lowest limit obtain in this way is 3.5Å. The value determined by the automatic procedure can be overruled by specification of the high_resolution keyword. The low resolution is set to 10Å by default.

Scope: parameters.reporting

```
keys: * verbose :: verbosity level.
* log :: log file name
* ccp4_style_graphs :: Either True or False. Determines whether or
not ccp4 style logfgra plots are written to the
log file
```

Scope: xray_data

```
keys: * file_name :: file name with xray data.
* obs_labels :: labels for observed data is format is mtz or XPLOR/CNS
* calc_labels :: optional; labels for calculated data
* unit_cell :: overrides unit cell in reflection file (if present)
* space_group :: overrides space group in reflection file (if present)
* high_resolution :: High resolution limit of the data
* low_resolution :: Low resolution limit of the data
```

Note that the matching of specified and present labels involves a sub-string matching algorithm.

Scope: optional

```
keys: * hklout :: output mtz file
* twinning.action :: Whether to detwin the data
* twinning.twin_law :: using this twin law (h,k,l or x,y,z notation)
* twinning.fraction :: The detwinning fraction.
* b_value :: the resulting Wilson B value
```

The output mtz file contains an anisotropy corrected mtz file, with suspected outliers removed. The data is put scaled and has the specified Wilson B value. These options have an associated expert level of 10, and are not shown by default. Specification of the expert level on the command line as 'level=100' will show all available options.

Interpreting Xtrriage output

Typing:

```
%phenix.xtriage some_data.sca residues=290 log=some_data.log
```

results in the following output (parts omitted).

Matthews analysis

First, a cell contents analysis is performed. Matthews coefficients, solvent content and solvent content probabilities are listed, and the most likely composition is guessed

Matthews coefficient and Solvent content statistics

Copies	Solvent content	Matthews Coed.	P(solvent cont.)
1	0.705	4.171	0.241
2	0.411	2.085	0.750
3	0.116	1.390	0.009
Best guess : 2 copies in the asu			

Data strength

The next step, the strength of the data is gauged by determining the completeness of the in resolution bins after application of several I/sigI cut off values

Completeness and data strength analysis

The following table lists the completeness in various resolution ranges, after applying a I/sigI cut. Miller indices for which individual I/sigI values are larger than the value specified in the top row of the table, are retained, while other intensities are discarded. The resulting completeness profiles are an indication of the strength of the data.

Res. Range	I/sigI>1	I/sigI>2	I/sigI>3	I/sigI>5	I/sigI>10	I/sigI>15
19.87 - 7.98	96.4%	95.3%	94.5%	93.6%	91.7%	89.3%
7.98 - 6.40	99.2%	98.2%	97.1%	95.5%	90.9%	84.7%
6.40 - 5.61	97.8%	95.4%	93.3%	87.1%	76.6%	66.8%
5.61 - 5.11	98.2%	95.9%	94.0%	87.9%	74.1%	58.0%
5.11 - 4.75	97.9%	96.2%	94.5%	91.1%	79.2%	62.5%
4.75 - 4.47	97.4%	95.4%	93.1%	88.9%	76.6%	56.9%
4.47 - 4.25	96.5%	94.5%	92.1%	88.0%	75.3%	56.5%
4.25 - 4.07	96.6%	94.0%	91.2%	85.4%	69.3%	44.9%
4.07 - 3.91	95.6%	92.1%	87.8%	80.1%	61.9%	34.8%
3.91 - 3.78	94.3%	89.6%	83.7%	71.1%	48.7%	20.5%
3.78 - 3.66	95.7%	90.9%	85.6%	71.5%	42.4%	14.8%
3.66 - 3.56	91.6%	85.0%	78.0%	63.3%	34.1%	9.5%
3.56 - 3.46	89.8%	80.4%	70.2%	52.8%	22.2%	3.8%
3.46 - 3.38	87.4%	76.3%	64.6%	46.7%	15.5%	1.7%

This analysis is also used in the automatic determination of the high resolution limit used in the intensity statistics and twin analyses.

Absolute, likelihood based Wilson scaling

The (anisotropic) B value of the data is determined using a likelihood based approach. The resulting B value/tensor is reported:

```
Maximum likelihood isotropic Wilson scaling
ML estimate of overall B value of sec17.sca:i_obs,sigma:
75.85 A**(-2)
```



```

Estimated -log of scale factor of sec17.sca:i_obs,sigma:
-2.50

Maximum likelihood anisotropic Wilson scaling
ML estimate of overall B_cart value of sec17.sca:i_obs,sigma:
68.92, 0.00, 0.00
68.92, 0.00
91.87
Equivalent representation as U_cif:
0.87, -0.00, -0.00
0.87, 0.00
1.16

ML estimate of -log of scale factor of sec17.sca:i_obs,sigma:
-2.50
Correcting for anisotropy in the data

```

A large spread in (especially the diagonal) values indicates anisotropy. The anisotropy is corrected for. This clears up intensity statistics.

Low resolution completeness analysis

Mostly data processing software do not provide a clear picture of the completeness of the data at low resolution. For this reason, xtriage lists the completeness of the data up to 5 Angstrom:

```

Low resolution completeness analysis

The following table shows the completeness
of the data to 5 Angstrom.
unused: - 19.8702 [ 0/68 ] 0.000
bin 1: 19.8702 - 10.3027 [425/455] 0.934
bin 2: 10.3027 - 8.3766 [443/446] 0.993
bin 3: 8.3766 - 7.3796 [446/447] 0.998
bin 4: 7.3796 - 6.7336 [447/449] 0.996
bin 5: 6.7336 - 6.2673 [450/454] 0.991
bin 6: 6.2673 - 5.9080 [428/429] 0.998
bin 7: 5.9080 - 5.6192 [459/466] 0.985
bin 8: 5.6192 - 5.3796 [446/450] 0.991
bin 9: 5.3796 - 5.1763 [437/440] 0.993
bin 10: 5.1763 - 5.0006 [460/462] 0.996
unused: 5.0006 - [ 0/0 ]

```

This analysis allows one to quickly see if there is any unusually low completeness at low resolution, for instance due to missing overloads.

Wilson plot analysis

A Wilson plot analysis a la ARP/wARP is carried out, albeit with a slightly different standard curve:

```

Mean intensity analysis
Analysis of the mean intensity.
Inspired by: Morris et al. (2004). J. Synch. Rad.11, 56-59.
The following resolution shells are worrisome:
-----
| d_spacing | z_score | compl. | <Iobs>/<Iexp> |
-----
| 5.773 | 7.95 | 0.99 | 0.658 |
| 5.423 | 8.62 | 0.99 | 0.654 |
| 5.130 | 6.31 | 0.99 | 0.744 |
| 4.879 | 5.36 | 0.99 | 0.775 |
| 4.662 | 4.52 | 0.99 | 0.803 |
| 3.676 | 5.45 | 0.99 | 1.248 |
-----

```

Possible reasons for the presence of the reported unexpected low or elevated mean intensity in a given resolution bin are :

- missing overloaded or weak reflections
- suboptimal data processing
- satellite (ice) crystals
- NCS
- translational pseudo symmetry (detected elsewhere)
- outliers (detected elsewhere)
- ice rings (detected elsewhere)
- other problems

Note that the presence of abnormalities in a certain region of reciprocal space might confuse the data validation algorithm throughout a large region of reciprocal space, even though the data is acceptable in those areas.

A very long list of warnings could indicate a serious problem with your data. Decisions on whether or not the data is useful, should be cut or should be thrown away altogether, is not straightforward and falls beyond the scope of xtriage.

Outlier detection and rejection

Possible outliers are detected on the basis Wilson statistics:

Possible outliers

Inspired by: Read, Acta Cryst. (1999). D55, 1759-1764

Acentric reflections:

```
-----
| d_space | H K L | |E| | p(wilson) | p(extreme) |
-----
| 3.716 | 8, 6, 31 | 3.52 | 4.06e-06 | 5.87e-02 |
-----
```

$p(\text{wilson}) : 1 - (1 - \exp[-|E|^2])$

$p(\text{extreme}) : 1 - (1 - \exp[-|E|^2])^{(n_{\text{acentrics}})}$

$p(\text{wilson})$ is the probability that an E-value of the specified value would be observed when it would be selected at random from the given data set.

$p(\text{extreme})$ is the probability that the largest $|E|$ value is larger or equal than the observed largest $|E|$ value.

Both measures can be used for outlier detection. $p(\text{extreme})$ takes into account the size of the data set.

Outliers are removed from the data set in the further analysis. Note that if pseudo translational symmetry is present, a large number of 'outliers' will be present.

Ice ring detection

Ice rings in the data are detected by analyzing the completeness and the mean intensity:

Ice ring related problems

The following statistics were obtained from ice-ring insensitive resolution ranges

mean bin z_score : 3.47

(rms deviation : 2.83)

mean bin completeness : 0.99

(rms deviation : 0.00)

The following table shows the z-scores and completeness in ice-ring sensitive areas. Large z-scores and high completeness in these resolution ranges might be a reason to re-assess your data processing if ice rings were present.

d_spacing	z_score	compl.	Rel. Ice int.
3.897	0.12	0.97	1.000
3.669	0.96	0.95	0.750
3.441	2.14	0.94	0.530

Abnormalities in mean intensity or completeness at resolution ranges with a relative ice ring intensity lower than 0.10 will be ignored.

At 3.67 Å there is an lower occupancy then expected from the rest of the data set. Even though the completeness is lower as expected, the mean intensity is still reasonable at this resolution

At 3.44 Å there is an lower occupancy then expected from the rest of the data set. Even though the completeness is lower as expected, the mean intensity is still reasonable at this resolution

There were 2 ice ring related warnings
This could indicate the presence of ice rings.

Anomalous signal

If the input reflection file contains separate intensities for each Friedel mate, a quality measure of the anomalous signal is reported:

Analysis of anomalous differences

Table of measurability as a function of resolution

The measurability is defined as the fraction of Bijvoet related intensity differences for which $|\Delta I|/\sigma_{\Delta I} \geq 3.0$
 $\min[I(+)/\sigma_{I(+)}, I(-)/\sigma_{I(-)}] \geq 3.0$ holds.

The measurability provides an intuitive feeling of the quality of the data, as it is related to the number of reliable Bijvoet differences. When the data is processed properly and the standard deviations have been estimated accurately, values larger than 0.05 are encouraging.

```
unused: - 19.8704 [ 0/68 ]
bin 1: 19.8704 - 7.0211 [1551/1585] 0.1924
bin 2: 7.0211 - 5.6142 [1560/1575] 0.0814
bin 3: 5.6142 - 4.9168 [1546/1555] 0.0261
bin 4: 4.9168 - 4.4729 [1563/1582] 0.0081
bin 5: 4.4729 - 4.1554 [1557/1577] 0.0095
bin 6: 4.1554 - 3.9124 [1531/1570] 0.0083
bin 7: 3.9124 - 3.7178 [1541/1585] 0.0069
bin 8: 3.7178 - 3.5569 [1509/1552] 0.0028
bin 9: 3.5569 - 3.4207 [1522/1606] 0.0085
bin 10: 3.4207 - 3.3032 [1492/1574] 0.0044
unused: 3.3032 - [ 0/0 ]
```

The anomalous signal seems to extend to about 5.9 Å (or to 5.2 Å, from a more optimistic point of view). The quoted resolution limits can be used as a guideline to decide where to cut the resolution for phenix.hyss. As the anomalous signal is not very strong in this data set, substructure solution via SAD might prove to be a challenge. Especially if only low resolution reflections are used, the resulting substructures could contain a significant amount of false positives.

Determination of twin laws

Twin laws are found using a modified le-Page algorithm and classified as merohedral and pseudo merohedral:

Determining possible twin laws.

The following twin laws have been found:

Type	Axis	R metric (%)	delta (le Page)	delta (Lebedev)	Twin law
M	2-fold	0.000	0.000	0.000	-h,k,-l

M: Merohedral twin law

PM: Pseudomerohedral twin law

1 merohedral twin operators found

0 pseudo-merohedral twin operators found

In total, 1 twin operator were found

Non-merohedral (reticular) twinning is not considered. The R-metric is equal to :

$$\text{Sum } (M_i - N_i)^2 / \text{Sum } M_i^2$$

M_i are elements of the original metric tensor and N_i are elements of the metric tensor after 'idealizing' the unit cell, in compliance with the restrictions the twin law poses on the lattice if it would be a 'true' symmetry operator.

The delta le-Page is the familiar obliquity. The delta Lebedev is a twin law quality measure developed by A. Lebedev (Lebedev, Vagin & Murshudov; Acta Cryst. (2006). D62, 83-95.).

Note that for merohedral twin laws, all quality indicators are 0. For non-merohedral twin laws, this value is larger or equal to zero. If a twin law is classified as non-merohedral, but has a delta le-page equal to zero, the twin law is sometimes referred to as a metric merohedral twin law.

Locating translational pseudo symmetry (TPS)

TPS is located by inspecting a low resolution Patterson function. Peaks and their significance levels are reported:

Largest Patterson peak with length larger then 15 Angstrom

Frac. coord. : 0.027 0.057 0.345

Distance to origin : 17.444

Height (origin=100) : 3.886

p_value(height) : 9.982e-01

The reported p_value has the following meaning:

The probability that a peak of the specified height or larger is found in a Patterson function of a macro molecule that does not have any translational pseudo symmetry is equal to 9.982e-01
p_values smaller then 0.05 might indicate weak translation pseudo symmetry, or the self vector of a large anomalous scatterer such as Hg, whereas values smaller then 1e-3 are a very strong indication for the presence of translational pseudo symmetry.

Moments of the observed intensities

The moment of the observed intensity/amplitude distribution, are reported, as well as their expected values:

Wilson ratio and moments

Acentric reflections
 $\langle I^2 \rangle / \langle I \rangle^2$: 1.955 (untwinned: 2.000; perfect twin 1.500)
 $\langle F^2 \rangle / \langle F \rangle^2$: 0.796 (untwinned: 0.785; perfect twin 0.885)
 $\langle |E|^2 - 1 \rangle$: 0.725 (untwinned: 0.736; perfect twin 0.541)

Centric reflections
 $\langle I^2 \rangle / \langle I \rangle^2$: 2.554 (untwinned: 3.000; perfect twin 2.000)
 $\langle F^2 \rangle / \langle F \rangle^2$: 0.700 (untwinned: 0.637; perfect twin 0.785)
 $\langle |E|^2 - 1 \rangle$: 0.896 (untwinned: 0.968; perfect twin 0.736)

Significant departure from the ideal values could indicate the presence of twinning or pseudo translations. For instance, an $\langle I^2 \rangle / \langle I \rangle^2$ value significantly lower than 2.0, might point to twinning, whereas a value significantly larger than 2.0, might point towards pseudo translational symmetry.

Cumulative intensity distribution

The cumulative intensity distribution is reported:

Z	Nac_obs	Nac_theo	Nc_obs	Nc_theo
0.0	0.000	0.000	0.000	0.000
0.1	0.081	0.095	0.168	0.248
0.2	0.167	0.181	0.292	0.345
0.3	0.247	0.259	0.354	0.419
0.4	0.321	0.330	0.420	0.474
0.5	0.392	0.394	0.473	0.520
0.6	0.452	0.451	0.521	0.561
0.7	0.506	0.503	0.570	0.597
0.8	0.552	0.551	0.603	0.629
0.9	0.593	0.593	0.636	0.657
1.0	0.635	0.632	0.673	0.683
Maximum deviation acentric : 0.015				
Maximum deviation centric : 0.080				
$\langle NZ(\text{obs}) - NZ(\text{twinned}) \rangle_{\text{acentric}}$: -0.004				
$\langle NZ(\text{obs}) - NZ(\text{twinned}) \rangle_{\text{centric}}$: -0.039				

The N(Z) test is related to the moments based test discussed above. Nac_obs is the observed cumulative distribution of normalized intensities of the acentric data, and uses the full distribution rather than just a moment.

The effects of twinning shows itself for Nac_obs having a more sigmoidal character. In the case of pseudo centering, Nac_obs will tend towards Nc_theo.

The L test

The L-test is an intensity statistic developed by Padilla and Yeates (Acta Cryst. (2003), D59: 1124-1130) and is reasonably robust in the presence of anisotropy and pseudo centering, especially if the miller indices are partitioned properly. Partitioning is carried out on the basis of a Patterson analysis. A significant deviation of both $\langle |L| \rangle$ and $\langle L^2 \rangle$ from the expected values indicate twinning or other problems:

L test for acentric data

using difference vectors (dh,dk,dl) of the form:
(2hp,2kp,2lp)
where hp, kp, and lp are random signed integers such that
 $2 \leq |dh| + |dk| + |dl| \leq 8$

Mean $|L|$: 0.482 (untwinned: 0.500; perfect twin: 0.375)
Mean L^2 : 0.314 (untwinned: 0.333; perfect twin: 0.200)

The distribution of $|L|$ values indicates a twin fraction of

0.00. Note that this estimate is not as reliable as obtained via a Britton plot or H-test if twin laws are available.

Whether or not the $\langle |L| \rangle$ and $\langle L^2 \rangle$ differ significantly from the expected values, is shown in the final summary (see below).

Analysis of twin laws

Twin law specific tests (Britton, H and RvsR) are performed:

Results of the H-test on a-centric data:

(Only 50.0% of the strongest twin pairs were used)

mean $|H|$: 0.183 (0.50: untwinned; 0.0: 50% twinned)
mean H^2 : 0.055 (0.33: untwinned; 0.0: 50% twinned)
Estimation of twin fraction via mean $|H|$: 0.317
Estimation of twin fraction via cum. dist. of H : 0.308

Britton analysis

Extrapolation performed on 0.34 < alpha < 0.495
Estimated twin fraction: 0.283
Correlation: 0.9951

R vs R statistic:

$R_{\text{abs_twin}} = \frac{|I_1 - I_2|}{|I_1 + I_2|}$
Lebedev, Vagin, Murshudov. Acta Cryst. (2006). D62, 83-95

$R_{\text{abs_twin}}$ observed data : 0.193
 $R_{\text{abs_twin}}$ calculated data : 0.328

$R_{\text{sq_twin}} = \frac{(I_1 - I_2)^2}{(I_1 + I_2)^2}$
 $R_{\text{sq_twin}}$ observed data : 0.044
 $R_{\text{sq_twin}}$ calculated data : 0.120

Maximum Likelihood twin fraction determination

Zwart, Read, Grosse-Kunstleve & Adams, to be published.

The estimated twin fraction is equal to 0.227

These tests allow one to estimate the twin fraction and (if calculated data is provided) determine if rotational pseudo symmetry is present. Another option (albeit more computationally expensive), is to estimate the correlation between error free, untwinned, twin related normalized intensities (use the key perform=True on the command line)

Estimation of twin fraction, while taking into account the effects of possible NCS parallel to the twin axis.
Zwart, Read, Grosse-Kunstleve & Adams, to be published.

A parameters D_{ncs} will be estimated as a function of resolution, together with a global twin fraction.
 D_{ncs} is an estimate of the correlation coefficient between untwinned, error-free, twin related, normalized intensities. Large values (0.95) could indicate an incorrect point group. Value of D_{ncs} larger than say, 0.5, could indicate the presence of NCS. The twin fraction should be smaller or similar to other estimates given elsewhere.

The refinement can take some time.
For numerical stability issues, D_{ncs} is limited between 0 and 0.95.
The twin fraction is allowed to vary between 0 and 0.45.
Refinement cycle numbers are printed out to keep you entertained.

```
. . . . 5 . . . . 10 . . . . 15 . . . . 20 . . . . 25 . . . . 30
. . . . 35 . . . . 40 . . . . 45 . . . . 50 . . . . 55 . . . . 60
```

. . . . 65 70 75 . . .

Cycle : 78

Log[likelihood]: 22853.700

twin fraction: 0.201

D_ncs in resolution ranges:

9.8232 -- 4.5978 :: 0.830

4.5978 -- 3.7139 :: 0.775

3.7139 -- 3.2641 :: 0.745

3.2641 -- 2.9747 :: 0.746

2.9747 -- 2.7666 :: 0.705

2.7666 -- 2.6068 :: 0.754

2.6068 -- 2.4784 :: 0.735

The correlation of the calculated F^2 should be similar to the estimated values.

Observed correlation between twin related, untwinned calculated F^2 in resolution ranges, as well as estimates D_{ncs}^2 values:

Bin d_max d_min CC_obs D_{ncs}^2

1) 9.8232 -- 4.5978 :: 0.661 0.689

2) 4.5978 -- 3.7139 :: 0.544 0.601

3) 3.7139 -- 3.2641 :: 0.650 0.556

4) 3.2641 -- 2.9747 :: 0.466 0.557

5) 2.9747 -- 2.7666 :: 0.426 0.497

6) 2.7666 -- 2.6068 :: 0.558 0.569

7) 2.6068 -- 2.4784 :: 0.531 0.540

The twin fraction obtained via this method is usually lower than what is obtained by refinement. The estimated correlation coefficient (D_{ncs}^2) between the twin related F^2 values, is however reasonably accurate.

Exploring higher metric symmetry

The fact that a twin law is present, could indicate that the data was incorrectly processed as well. The example below, shows a P41212 data set processed in P1:

Exploring higher metric symmetry

Point group of data as dictated by the space group is P 1

the point group in the Niggli setting is P 1

The point group of the lattice is P 4 2 2

A summary of R values for various possible point groups follow.

| Point group | mean R_used | max R_used | mean R_unused | min R_unused | choice |

P 1	None	None	0.022	0.017	
P 4 2 2	0.022	0.025	None	None	<---
P 1 2 1	0.017	0.017	0.026	0.024	
Hall: C 2y (x-y,x+y,z)	0.025	0.025	0.022	0.017	
P 4	0.025	0.028	0.025	0.025	
Hall: C 2 2 (x-y,x+y,z)	0.024	0.025	0.017	0.017	
Hall: C 2y (x+y,-x+y,z)	0.024	0.024	0.023	0.017	
P 1 1 2	0.028	0.028	0.021	0.017	
P 2 1 1	0.027	0.027	0.022	0.017	
P 2 2 2	0.023	0.028	0.025	0.025	

R_used: mean and maximum R value for symmetry operators *used* in this point group

R_unused: mean and minimum R value for symmetry operators *not used* in this point group

The likely point group of the data is: P 4 2 2

As in phenix.explore_metric_symmetry, the possible space groups are listed as well (not shown here).

Twin analysis summary

The results of the twin analysis are summarized. Typical outputs look as follows for cases of wrong symmetry, twin laws but no suspected twinning and twinned data respectively.

Wrong symmetry:

Twinning and intensity statistics summary (acentric data):

Statistics independent of twin laws

- $\langle I^2 \rangle / \langle I \rangle^2$: 2.104
- $\langle F \rangle^2 / \langle F^2 \rangle$: 0.770
- $\langle |E^2 - 1| \rangle$: 0.757
- $\langle |L| \rangle, \langle |L^2| \rangle$: 0.512, 0.349

Multivariate Z score L-test: 2.777

The multivariate Z score is a quality measure of the given spread in intensities. Good to reasonable data is expected to have a Z score lower than 3.5.

Large values can indicate twinning, but small values do not necessarily exclude it.

Statistics depending on twin laws

| Operator | type | R obs. | Britton alpha | H alpha |

k,h,-l	PM	0.025	0.458	0.478
-h,k,-l	PM	0.017	0.459	0.487
-k,h,l	PM	0.024	0.458	0.478
-k,-h,-l	PM	0.024	0.458	0.478
-h,-k,l	PM	0.028	0.458	0.476
h,-k,-l	PM	0.027	0.458	0.477
k,-h,l	PM	0.024	0.457	0.478

Patterson analysis

- Largest peak height : 6.089
(corresponding p value : 6.921e-01)

The largest off-origin peak in the Patterson function is 6.09% of the height of the origin peak. No significant pseudo-translation is detected.

The results of the L-test indicate that the intensity statistics behave as expected. No twinning is suspected.

The symmetry of the lattice and intensity however suggests that the input space group is too low. See the relevant sections of the log file for more details on your choice of space groups.

As the symmetry is suspected to be incorrect, it is advisable to reconsider data processing.

Twin laws present but no suspected twinning:

Twinning and intensity statistics summary (acentric data):

Statistics independent of twin laws

- $\langle I^2 \rangle / \langle I \rangle^2$: 1.955
- $\langle F \rangle^2 / \langle F^2 \rangle$: 0.796
- $\langle |E^2 - 1| \rangle$: 0.725
- $\langle |L| \rangle, \langle |L^2| \rangle$: 0.482, 0.314

Multivariate Z score L-test: 1.225

The multivariate Z score is a quality measure of the given spread in intensities. Good to reasonable data is expected to have a Z score lower than 3.5.

Large values can indicate twinning, but small values do not necessarily exclude it.

Statistics depending on twin laws


```

-----
| Operator | type | R obs. | Britton alpha | H alpha |
-----
| -h,k,-l | M | 0.455 | 0.016 | 0.035 |
-----

```

Patterson analysis

- Largest peak height : 3.886
(corresponding p value : 9.982e-01)

The largest off-origin peak in the Patterson function is 3.89% of the height of the origin peak. No significant pseudo-translation is detected.

The results of the L-test indicate that the intensity statistics behave as expected. No twinning is suspected.

Even though no twinning is suspected, it might be worthwhile carrying out a refinement using a dedicated twin target anyway, as twinned structures with low twin fractions are difficult to distinguish from non-twinned structures.

Twinned data:

```

-----
Twinning and intensity statistics summary (acentric data):

```

Statistics independent of twin laws

- $\langle I^2 \rangle / \langle I \rangle^2$: 1.587
- $\langle F^2 \rangle / \langle F \rangle^2$: 0.871
- $\langle |E^2 - 1| \rangle$: 0.568
- $\langle |L| \rangle, \langle |L^2| \rangle$: 0.387, 0.212

Multivariate Z score L-test: 11.589

The multivariate Z score is a quality measure of the given spread in intensities. Good to reasonable data is expected to have a Z score lower than 3.5.

Large values can indicate twinning, but small values do not necessarily exclude it.

Statistics depending on twin laws

```

-----
| Operator | type | R obs. | Britton alpha | H alpha |
-----
| -l,-k,-h | PM | 0.170 | 0.330 | 0.325 |
-----

```

Patterson analysis

- Largest peak height : 7.300
(corresponding p value : 4.454e-01)

The largest off-origin peak in the Patterson function is 7.30% of the height of the origin peak. No significant pseudo-translation is detected.

The results of the L-test indicate that the intensity statistics are significantly different than is expected from good to reasonable, untwinned data.

As there are twin laws possible given the crystal symmetry, twinning could be the reason for the departure of the intensity statistics from normality. It might be worthwhile carrying refinement with a twin specific target function.

In the summary, the significance of the departure of the values of the L-test from normality are stated. The multivariate Z-score (also known as the Mahalanobis distance) is used for this purpose.

Examples

Standard run of xtriage

Running xtriage is easy. From the command-line you can type:

```
phenix.xtriage data.sca
```

When an MTZ or CNS file is used, labels have to be specified:

```
phenix.xtriage file=my_brilliant_data.mtz obs_labels='F(+),SIGF(+),F(-),SIGF(-)'
```

In order to perform a Matthews analysis, it might be useful to specify the number of residues/nucleotides in the crystallized macro molecule:

```
phenix.xtriage data.sca n_residues=230 n_bases=25
```

By default, the screen output plus additional ccp4 style graphs (viewable with the ccp4 programs loggraph) are echoed to a file named logfile.log. The command line arguments and all other defaults settings are summarized in a PHIL parameter data block given at the beginning of the logfile / screen output:

```
scaling.input {
  parameters {
    asu_contents {
      n_residues = None
      n_bases = None
      n_copies_per_asu = None
    }
    misc_twin_parameters {
      missing_symmetry {
        tanh_location = 0.08
        tanh_slope = 50
      }
      twinning_with_ncs {
        perform_analysis = False
        n_bins = 7
      }
      twin_test_cuts {
        low_resolution = 10
        high_resolution = None
        isigi_cut = 3
        completeness_cut = 0.85
      }
    }
    reporting {
      verbose = 1
      log = "logfile.log"
      ccp4_style_graphs = True
    }
  }
  xray_data {
    file_name = "some_data.sca"
    obs_labels = None
    calc_labels = None
    unit_cell = 64.5 69.5 45.5 90 104.3 90
    space_group = "P 1 21 1"
    high_resolution = None
    low_resolution = None
  }
}
```

The defaults are good for most applications.

Possible Problems

Specific limitations and problems

- Xtriage doesn't deal with data in centric space groups

Literature

CCP4 newsletter No. 42, Summer 2005: Characterization of X-ray data sets CCP4 newsletter No. 43, Winter 2005: Xtriage and Fest: automatic assessment of X-ray data and substructure structure factor estimation

- CCP4 newsletter No. 42, Summer 2005: Characterization of X-ray data sets
- CCP4 newsletter No. 43, Winter 2005: Xtriage and Fest: automatic assessment of X-ray data and substructure structure factor estimation

Additional information

List of all available keywords

scalinginputexpert_level = 1 Expert level
 asu_contents Defines the ASU contents
 sequence_file = None File containing protein or nucleic acid sequences. Values for n_residues and n_bases will be extracted automatically if this is provided.
 n_residues = None Number of residues in structural unit
 n_bases = None Number of nucleotides in structural unit
 n_sites = 5 Number of atoms in anomalous substructure
 n_copies_per_asu = None Number of copies per ASU. If not specified, Matthews analyses is performed
 xray_data Defines xray datafile_name = None File name with data
 obs_labels = None Labels for observed data
 calc_labels = None Labels for calculated data
 unit_cell = None Unit cell parameters
 space_group = None space group
 high_resolution = None High resolution limit
 low_resolution = None Low resolution limit
 skip_sanity_checks = False Disable guards against corrupted input
 data_completeness_as_non_anomalous = True Report completeness of non-anomalous version of arrays
 reference A reference data set. For the investigation of possible reindexing options
 data Defines an x-ray dataset
 file_name = None File name
 labels = None Labels
 unit_cell = None Unit cell parameters
 space_group = None Space group
 structurefile_name = None Filename of reference PDB file
 parameters Basic settings
 reporting Some output issues
 verbose = 1 Verbosity
 log = logfile.log Logfile
 loggraphs = False merging
 n_bins = 10 skip_merging = False misc_twin_parameters Various settings for twinning or symmetry tests
 l_test_dhkl = None Offsets for hkl in applying L-test. Defaults to 2,2,2 but should be changed for commensurate modulation (tNCS) involving shifts other than 1/2 along x,y,z
 apply_basic_filters_prior_to_twin_analysis = True Keep data cutoffs from the basic_analyses module (1/sigma, Wilson scaling, Anisotropy) when twin stats are computed.
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 sigma_inflation = 1.25 Standard deviations of intensities can be increased to make point group determination more reliable.
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 n_bins = 7 Number of bins used in NCS analyses.
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 low_resolution = 10.0 Low resolution
 high_resolution = None High resolution
 isigi_cut = 3.0 I/sigI ratio used in completeness cut
 completeness_cut = 0.85 Data is cut at resolution where intensities with I/sigI greater than isigi_cut are more than completeness_cut complete
 optional_hklout = None hklout_type = mtz sca *Auto
 label_extension = "massaged" aniso Parameters dealing with anisotropy correction
 action = *remove_aniso None Remove anisotropy?
 final_b = *eigen_min eigen_mean user_b_iso Final b value
 b_iso = None User specified B value
 outlier Outlier analyses
 action = *extreme basic beamstop None Outlier protocol
 parameters Parameters for outlier detection
 basic_wilsonlevel = 1E-6 extreme_wilsonlevel = 0.01 beamstoplevel = 0.001
 d_min = 10.0 symmetry action = detwin twin *None twinning_parameterstwin_law = None
 fraction = None gui GUI-specific parameters, not applicable to command-line version.
 result_file = None Pickled result file for Phenix GUI
 output_dir = None job_title = None Job title in PHENIX GUI, not used on command line

- scalinginputexpert_level = 1 Expert levelasu_contentsDefines the ASU contentssequence_file = None File containing protein or nucleic acid sequences. Values for n_residues and n_bases will be extracted automatically if this is provided.n_residues = None Number of residues in structural unitn_bases = None Number of nucleotides in structural unitn_sites = 5 Number of atoms in anomalous substructureen_copies_per_asu = None Number of copies per ASU. If not specified, Matthews analyses is performedxray_dataDefines xray datafile_name = None File name with dataobs_labels = None Labels for observed datacalc_labels = None Labels for calculated dataunit_cell = None Unit cell parametersspace_group = None space grouphigh_resolution = None High resolution limitlow_resolution = None Low resolution limitsskip_sanity_checks = False Disable guards against corrupted input data.completeness_as_non_anomalous = True Report completeness of non-anomalous version of arraysreferenceA reference data set. For the investigation of possible reindexing optionsdataDefines an x-ray datasetfile_name = None File namelabels = None Labelsunit_cell = None Unit cell parameters"space_group = None Space groupstructurefile_name = None Filename of reference PDB fileparametersBasic settingsreportingSome output issuesverbose = 1 Verbositylog = logfile.log Logfileloggraphs = False mergingn_bins = 10 skip_merging = False misc_twin_parametersVarious settings for twinning or symmetry testsl_test_dhkl = None Offsets for hkl in applying L-test. Defaults to 2,2,2 but should be changed for commensurate modulation (tNCS) involving shifts other than 1/2 along x,y,zapply_basic_filters_prior_to_twin_analysis = True Keep data cutoffs from the basic_analyses module (l/sigma,Wilson scaling,Anisotropy) when twin stats are computed.missing_symmetrySettings for missing symmetry testssigma_inflation = 1.25 Standard deviations of intensities can be increased to make point group determination more reliable.twinning_with_ncsAnalysing the possibility of an NCS operator parallel to a twin law.perform_analyses = False Determines whether or not this analyses is carried out.n_bins = 7 Number of bins used in NCS analyses.twin_test_cutsVarious cuts used in determining resolution limit for data used in intensity statistics low_resolution = 10.0 Low resolutionhigh_resolution = None High resolutionisigi_cut = 3.0 l/sigl ratio used in completeness cut completeness_cut = 0.85 Data is cut at resolution where intensities with l/sigl greater than isigi_cut are more than completeness_cut completeoptionalhklout = None hklout_type = mtz sca *Auto label_extension = "massaged" anisoParameters dealing with anisotropy correctionaction = *remove_aniso None Remove anisotropy?final_b = *eigen_min eigen_mean user_b_iso Final b valueb_iso = None User specified B valueoutlierOutlier analysesaction = *extreme basic beamstop None Outlier protocolparametersParameters for outlier detectionbasic_wilsonlevel = 1E-6 extreme_wilsonlevel = 0.01 beamstoplevel = 0.001 d_min = 10.0 symmetryaction = detwin twin *None twinning_parameterstwin_law = None fraction = None guiGUI-specific parameters, not applicable to command-line version.result_file = None Pickled result file for Phenix GUIoutput_dir = None job_title = None Job title in PHENIX GUI, not used on command line

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 aniso Parameters dealing with anisotropy correction action = *remove_aniso None Remove
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 protocol parameters Parameters for outlier detection basic_wilson level = 1E-6 extreme_wilson level = 0.01
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- space_group = None space group
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- hklout_type = mtz sca *Auto
- label_extension = "massaged"
- anisoParameters dealing with anisotropy correctionaction = *remove_aniso None Remove anisotropy?final_b = *eigen_min eigen_mean user_b_iso Final b valueb_iso = None User specified B value
- action = *remove_aniso None Remove anisotropy?
- final_b = *eigen_min eigen_mean user_b_iso Final b value

- b_iso = None User specified B value
- outlierOutlier analysesaction = *extreme basic beamstop None Outlier protocolparametersParameters for outlier detectionbasic_wilsonlevel = 1E-6 extreme_wilsonlevel = 0.01 beamstoplevel = 0.001 d_min = 10.0
- action = *extreme basic beamstop None Outlier protocol
- parametersParameters for outlier detectionbasic_wilsonlevel = 1E-6 extreme_wilsonlevel = 0.01 beamstoplevel = 0.001 d_min = 10.0
- basic_wilsonlevel = 1E-6
- level = 1E-6
- extreme_wilsonlevel = 0.01
- level = 0.01
- beamstoplevel = 0.001 d_min = 10.0
- level = 0.001
- d_min = 10.0
- symmetryaction = detwin twin *None twinning_parameterstwin_law = None fraction = None
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- twinning_parameterstwin_law = None fraction = None
- twin_law = None
- fraction = None
- guiGUI-specific parameters, not applicable to command-line version.result_file = None Pickled result file for Phenix GUIoutput_dir = None job_title = None Job title in PHENIX GUI, not used on command line
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The phenix.xtriage graphical interface

xtriage_gui.html

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- Interpreting Xtriage output Alerts for problematic results Merging statistics Data quality analyses Low resolution completeness analysis. Wilson scaling. Ice ring detection. Anomalous signal. Twinning analyses Intensity distribution statistics Twin law tests
- Alerts for problematic results
- Merging statistics
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- Low resolution completeness analysis.
- Wilson scaling.
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- Anomalous signal.
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- Twin law tests

Video Tutorial

The basic xtriage tutorial video is available on the Phenix YouTube channel and explains how to run phenix.xtriage via the GUI.

The advanced tutorials cover (click on the link to access the video on the tutorial channel):

Translational NCS (tNCS) Twinning Space Group Identification, Ice rings & Wilson Plot

- Translational NCS (tNCS)
- Twinning
- Space Group Identification, Ice rings & Wilson Plot

Configuration

The minimum input for xtriage is a reflection file and symmetry information, which for MTZ files is automatically extracted. A sequence file is helpful for calculating expected solvent content. A model file (PDB or mmCIF), if available, will be used in the calculation of twinning statistics.

To launch Xtriage, open the main Phenix GUI and click on or drag-and-drop a reflection file onto the Xtriage module. The command-line equivalent is phenix.xtriage_gui; most parameters recognized by the command-line program should work here as well. The configuration window is minimal:

The Parameters button opens a window containing the most essential information about the reflection data, including choice of data to analyze if more than one suitable Miller array is present. Xtriage will accept any

reflection file format and both intensities and amplitudes, but a Scalepack or MTZ file with separate I+/- is the most typical input. (We recommend using unmerged data as it will be used in additional analyses of data quality and anomalous signal, but this is optional.) Most of this information will be extracted automatically if possible; at a minimum, only the input data and crystal symmetry are required. A complete listing of parameters can be found in the manual for the command-line version. Xtriage is also run automatically by the wizards (AutoSol, AutoBuild) and the graphical results can be viewed from the GUIs for those programs as well.

Interpreting Xtriage output

The output of the GUI is nearly identical in content to the output of the command-line version. The analyses performed are divided into several categories, covered in more detail below:

Merging statistics: these evaluate the agreement and quality of redundant experimental observations. Only included if unmerged data were used as input. Solvent content and Matthews coefficient: attempts to guess the crystal contents based on unit cell and space group (and optionally, information about the crystallized molecules). Data strength and completeness: analysis of signal-to-noise ratio both as an overall average and as a fraction of all possible reflections, and checking for pathologically incomplete data. Absolute scaling and Wilson analysis: calculates overall isotropic and anisotropic B-factors for the data, and compares the Wilson plot to an empirical function derived from PDB structures. This also includes the detection of ice rings and other pathologies. Anomalous signal: indicates approximate quality of anomalous signal, if anomalous data were provided. Systematic absences: examines the intensity of systematically absent reflections given different space group choices. (Note that if you have already scaled in a specific space group (e.g. P212121), these reflections may have already been removed from the input file.) Diagnostic tests for twinning and pseudosymmetry: analysis of intensity distribution statistics to detect abnormal data, usually the result of twinning or translational NCS. Twin law tests: determine which twin laws are permitted in the given crystal symmetry, and estimate the twin fraction in each case. (Note that these are purely theoretical guesses, and do not by themselves indicate that the data are actually twinned.) Exploring higher metric symmetry: if the crystal symmetry permits merging into higher-symmetry space groups, calculate the merging R-factors in each case.

- Merging statistics: these evaluate the agreement and quality of redundant experimental observations. Only included if unmerged data were used as input.
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indicate that the data are actually twinned.)

- Exploring higher metric symmetry: if the crystal symmetry permits merging into higher-symmetry space groups, calculate the merging R-factors in each case.

Several additional sections summarize the results from various analyses. You can navigate between sections via the drop-down menu near the top of the results tab.

Alerts for problematic results

The first tab provides warnings for a variety of common problems:

Each box can be clicked to jump to the detailed results in another tab. Analyses tagged with a red icon indicate issues that are likely to make various downstream steps such as phasing and refinement difficult (to varying degrees). Some of these may be remedied by better data processing, but they may be due to inherent issues of crystal quality. Yellow icons indicate issues that may produce sub-optimal results but are not necessarily fatal - for instance, if the resolution limit appears to have been chosen too conservatively. Green icons indicate that the data are of acceptable quality.

Merging statistics

If you provide unmerged (but scaled) intensities as input, the program `phenix.merging_statistics` will be run internally and the results displayed in their own tab.

In addition to the overall analysis, tables and plots display statistics by resolution shell, first for general counting statistics and signal-to-noise, second for various measures of data quality (including R-merge and CC1/2).

Clicking the plot button below a table opens plots for each set of analyses in a separate window.

Data quality analyses

Data strength. Xtriage displays two graphs of signal-to-noise ratio, one overall plot by resolution range, and a second combining signal-to-noise and completeness (shown below). These are very similar to the output of data-processing programs such as HKL2000 and the CCP4 suite (MOSFLM/SCALA), and usually do not vary greatly unless the data are systematically incomplete and/or very weak.

This analysis is also used in the automatic determination of the high resolution limit used in the intensity statistics and twin analyses. A separate table shows completeness for low-resolution data alone.

Low resolution completeness analysis.

Most data processing software do not provide a clear picture of the completeness of the data at low resolution. For this reason, xtriage lists the completeness of the data up to 5 Angstrom:

This analysis allows one to quickly see if there is any unusually low completeness at low resolution, for instance due to missing overloads.

Wilson scaling.

A Wilson plot analysis a la ARP/wARP is carried out, albeit with a slightly different standard curve. This determines overall B-factors (isotropic and anisotropic) for the data. A large spread in the diagonal values indicates anisotropy, as shown below (however, this example, the p9-sad structure included in the examples directory of the Phenix distribution, is actually very good data). The shape of the curve does not vary greatly

for most data, and a large deviation from expected values is unusual. However, the resolution at which the maximum of the curve is found will be different depending on the type of molecule crystallized (protein vs. nucleic acid).

Problems with the Wilson plot will be identified and reported; again, the p9-sad example is relatively free of pathologies:

A very long list of warnings could indicate a serious problem with your data. Decisions on whether or not the data is useful, should be cut or should be thrown away altogether, is not straightforward and falls beyond the scope of Xtriage.

Ice ring detection.

Ice rings in the data are detected by analyzing the completeness and the mean intensity:

Anomalous signal.

If the input reflection file contains separate intensities for each Friedel mate, an approximate measure of the anomalous signal is reported. The p9-sad example, a high-resolution SeMet SAD dataset, is shown again:

Suggested resolution cutoffs at which the anomalous signal disappears are shown by the vertical lines. Note that these are intended for heavy-atom substructure determination using for example HySS; they are not especially robust as a measure of quality. The p9-sad example is unusually excellent; a more difficult (but still solvable) dataset, the sec17 example in the Phenix distribution, is below:

Measurability values above 0.05 are encouraging; if the data do not reach this threshold at a usable resolution, successful phasing by SAD/MAD is extremely unlikely.

A more useful indicator of anomalous signal quality is the CC(anom), which is calculated if unmerged data are provided. This is essentially similar to CC1/2: the unmerged observations are separated into two pools, each of which is then merged (keeping Friedel mates separate), and the correlation coefficient of their anomalous differences is calculated. Values above 0.3 are very encouraging.

Twinning analyses

Xtriage will print a summary of its interpretation of twinning analyses at the top of the tab, including merohedral and pseudo-merohedral twin laws compatible with the lattice. Below is the output for the pseudo-merohedrally twinned porin-twin example data:

Intensity distribution statistics

Two separate tests for twinning rely on the cumulative distribution of intensities. Both of these result in plots with a distinctive shapes for normal vs. twinned data. The first, the NZ test, is shown below for the untwinned p9-sad data:

The plot for a twinned dataset has sigmoidal curve instead:

Translational NCS (and translational pseudosymmetry) will also result in an abnormal curve, shifted up relative to the expected distribution. Below is the NZ test plot for an untwinned dataset with translational pseudosymmetry:

Because the combination of twinning and translational NCS may result in a curve that appears close to normal, it is important to consider the second test for twinning, the L-test. Here, twinning is indicated by an upwards shift of the curve relative to expected values:

Twin law tests

If the crystal symmetry is compatible with one or more twin laws, each of these will be analyzed separately to determine the expected twin fraction. Most of these will result in a very low twin fraction whether or not twinning is present, but the actual twin law, if any, will usually have a significantly higher value, reflected in the shape of the curves. Results from an untwinned crystal typically look like this:

The actual twin law for the porin-twin example: